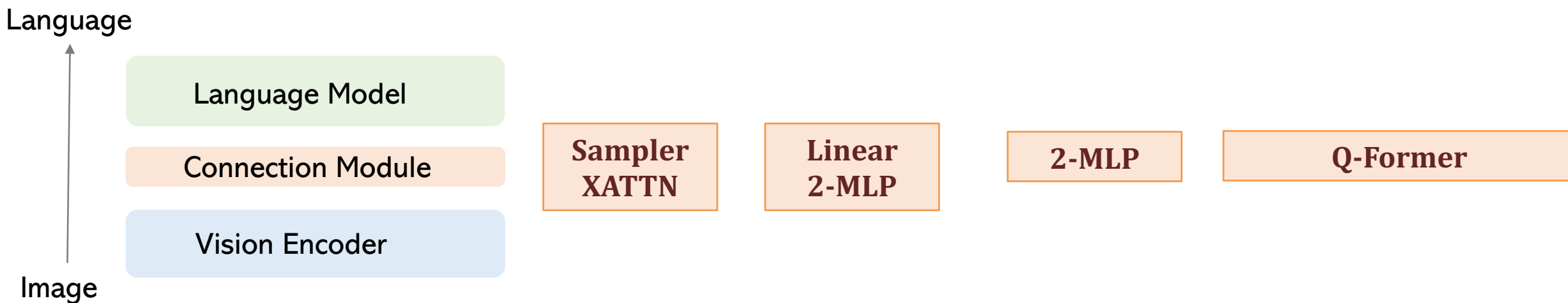
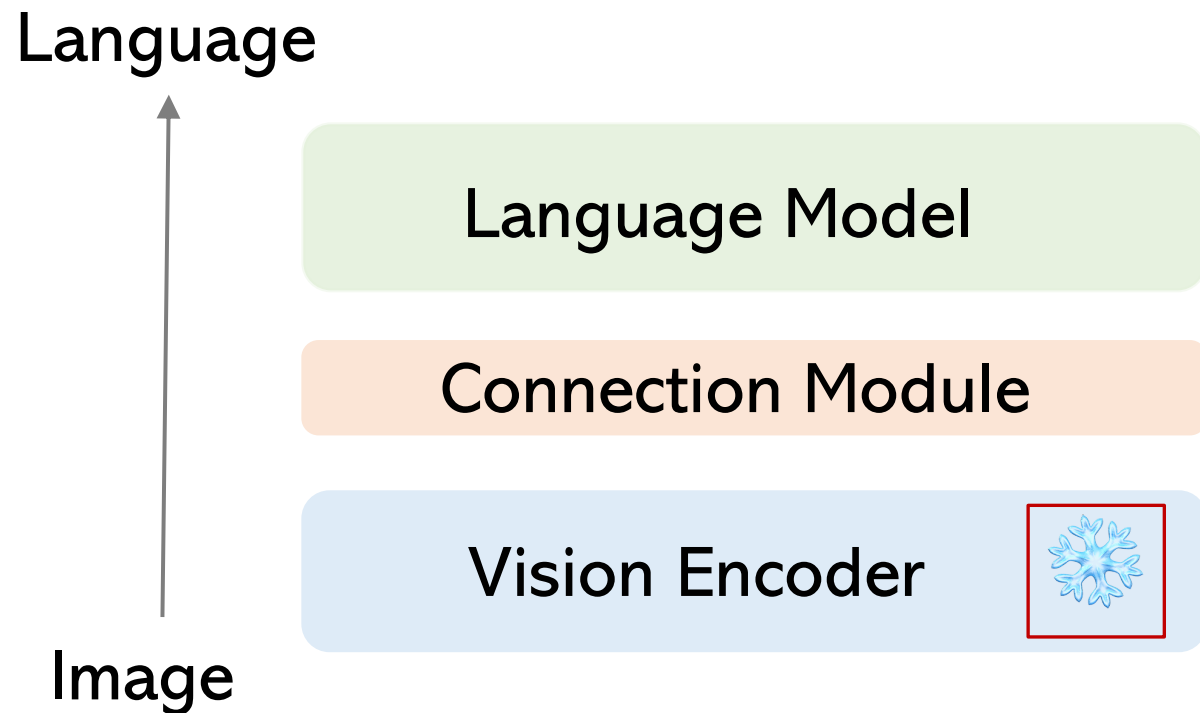


技术路线1

■ Flamingo ■ LLaVA ■ COGVLM ■ Qwen ■ BLIP*



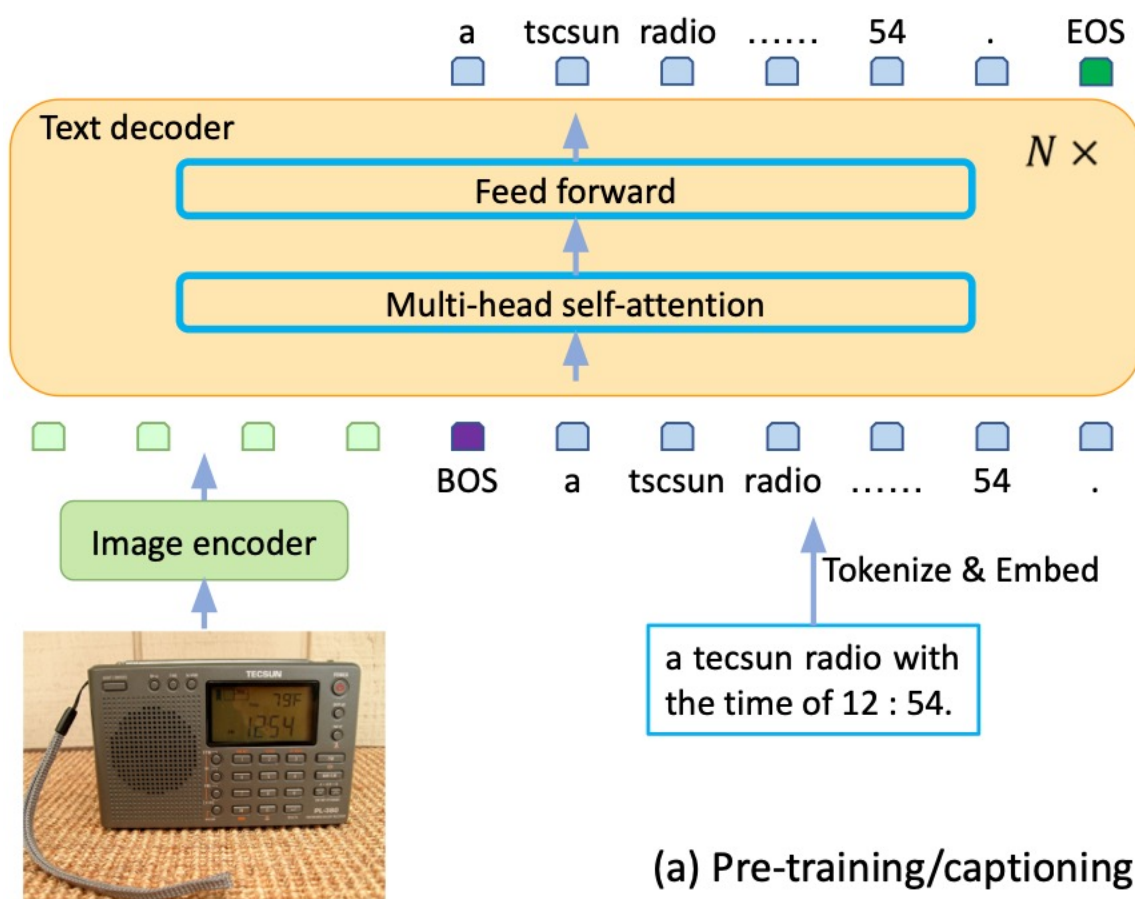


GIT: A Generative Image-to-text Transformer for Vision and Language

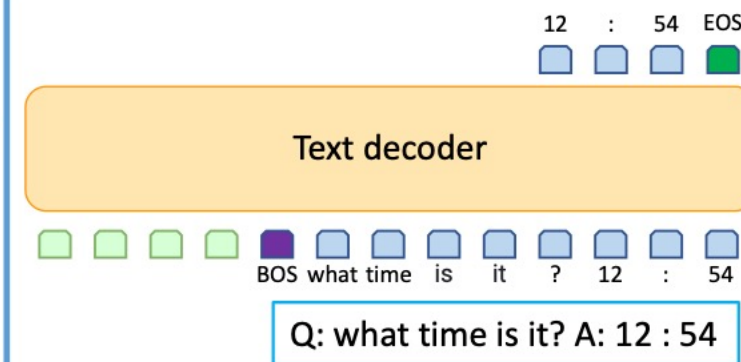
Jianfeng Wang
Zhengyuan Yang
Xiaowei Hu
Linjie Li
Kevin Lin
Zhe Gan
Zicheng Liu
Ce Liu
Lijuan Wang
Microsoft Cloud and AI

jianfw@microsoft.com
zhengyang@microsoft.com
xiaowei.hu@microsoft.com
lindsey.li@microsoft.com
keli@microsoft.com
zhe.gan@microsoft.com
zliu@microsoft.com
ce.liu@microsoft.com
lijuanw@microsoft.com

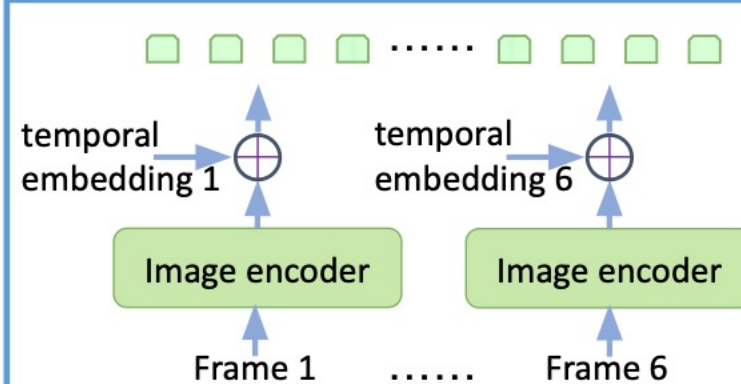
ViT-B/16
ViT-L/14
CoSwin
DaViT



(a) Pre-training/captioning



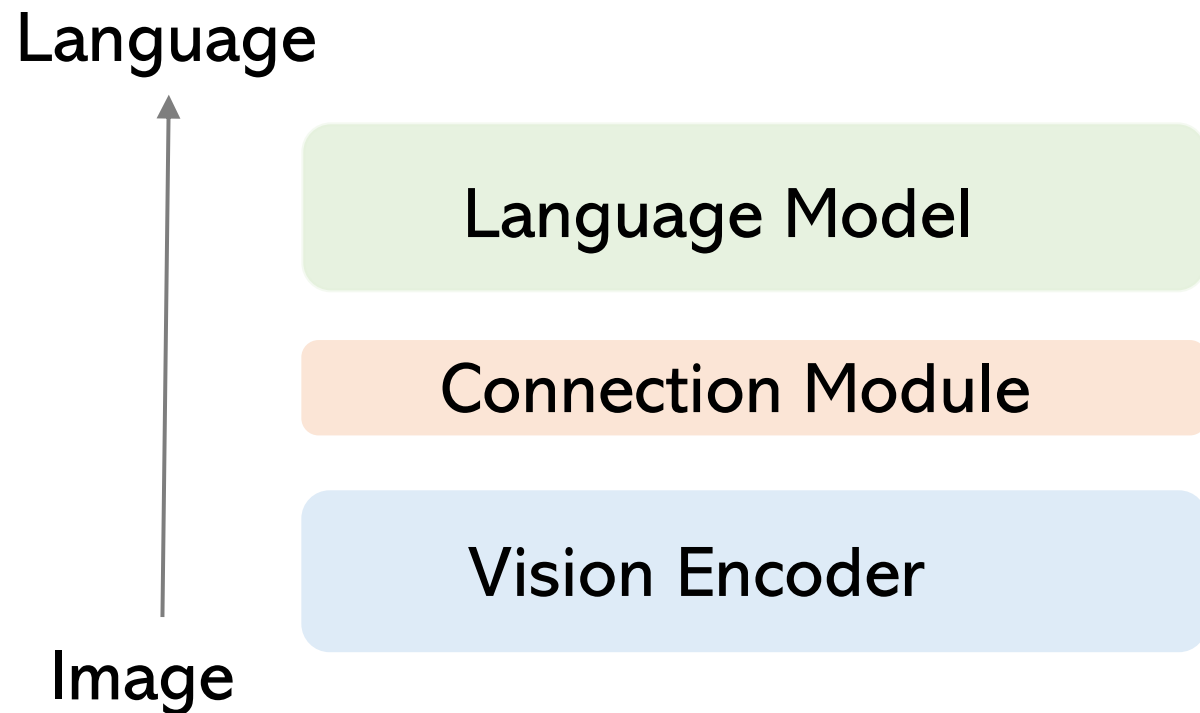
(b) VQA



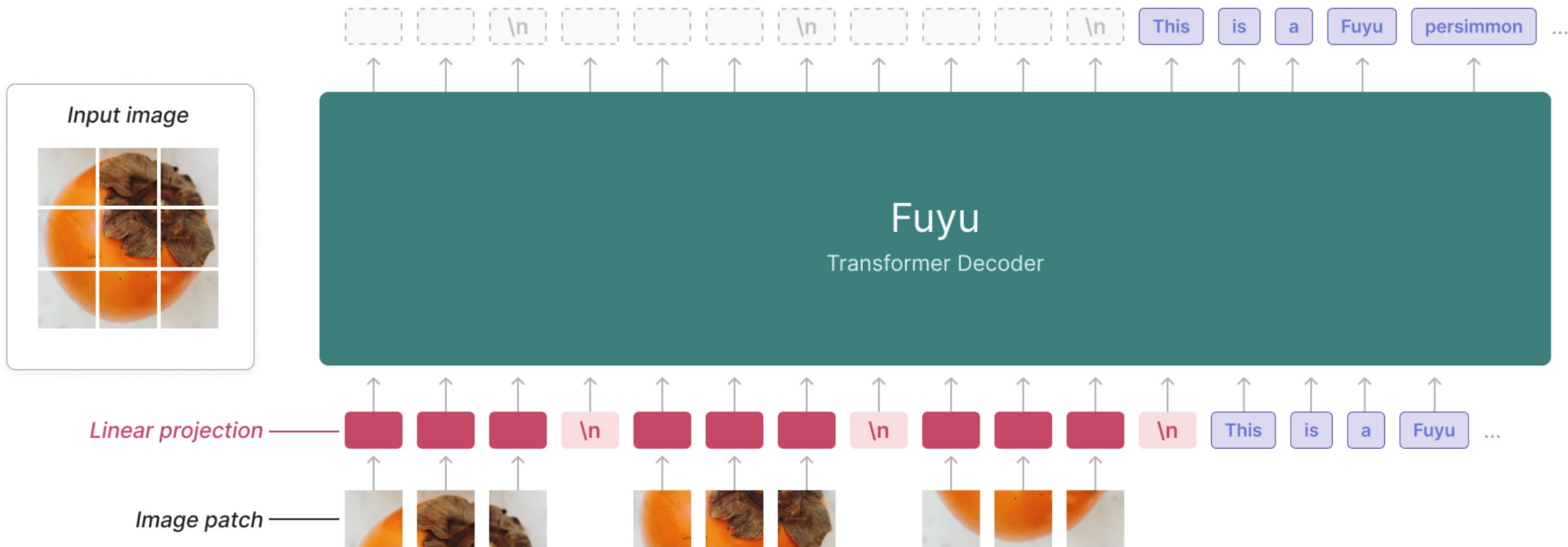
(c) Video

Name	images	image-text pairs	image encoder	epochs	model size	image size
GIT _B	4M	10M	CLIP/ViT-B/16 (Radford et al., 2021)	30	129M	224
GIT _L	14M	20M	CLIP/ViT-L/14 (Radford et al., 2021)	30	347M	224
GIT	0.8B	0.8B	Florence/CoSwin (Yuan et al., 2021)	2	681M	384
GIT2	10.5B	12.9B	DaViT (Ding et al., 2022) (4.8B)	2	5.1B	384

	Image captioning				Image QA			Video captioning				Video QA		Text Rec.
	COCO*	nocaps*	VizWiz*	TextCaps*	ST-VQA*	VizWiz*	OCR-VQA	MSVD	MSRVTT	VATEX*	TVC*	MSVD-QA	TGIF-Frame	Avg on 6
Prior SOTA ¹	138.7	120.6	94.1	109.7	69.6	65.4	67.9	120.6	60	86.5	64.5	48.3	69.5	93.8
GIT (ours)	148.8	123.4	114.4	138.2	69.6	67.5	68.1	180.2	73.9	93.8	61.2	56.8	72.8	92.9
Δ	+10.1	+2.8	+20.3	+28.5	+0.0	+2.1	+0.2	+59.6	+13.9	+7.3	-3.3	+8.5	+3.3	-0.9
GIT2 (ours)	149.8	124.8	120.8	145.0	75.8	70.1	70.3	185.4	75.9	96.6	65.0	58.2	74.9	94.5
Δ	+11.1	+ 4.2	+26.7	+35.3	+6.2	+4.7	+2.4	+64.8	+15.9	+10.1	+0.5	+9.9	+5.4	+0.7



Fuyu-8B



- 简化架构
- 任意尺寸与分辨率
- 统一的位置编码

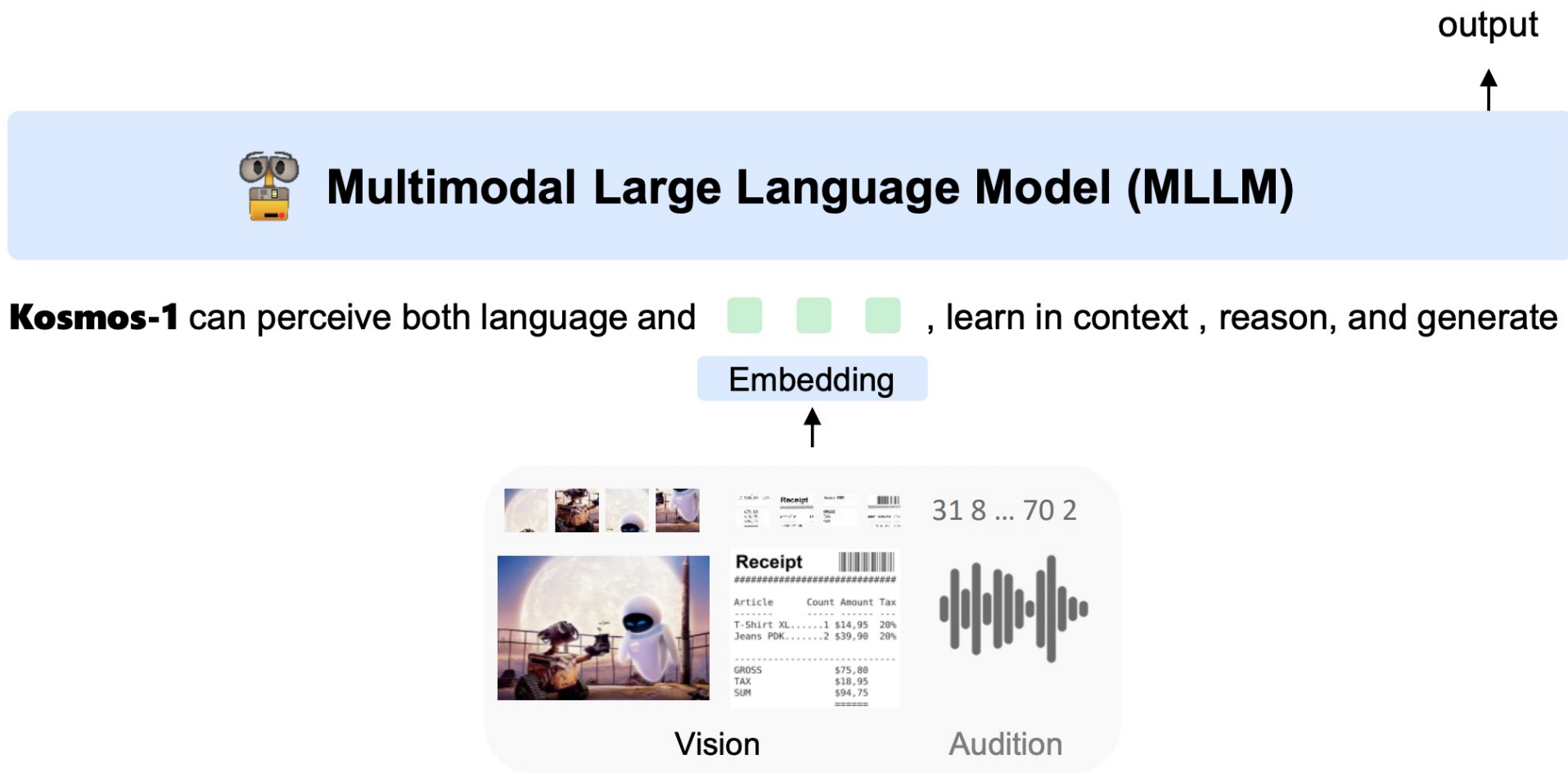
Fuyu-8B

Eval Task	Fuyu-8B	Fuyu-Medium	LLaVA 1.5 (13.5B)	QWEN-VL (10B)	PALI-X (55B)	PALM-e-12B	PALM-e-562B
VQAv2	74.2	77.4	80	79.5	86.1	76.2	80.0
OKVQA	60.6	63.1	n/a	58.6	66.1	55.5	66.1
COCO Captions	141	138	n/a	n/a	149	135	138
AI2D	64.5	73.7	n/a	62.3	81.2	n/a	n/a

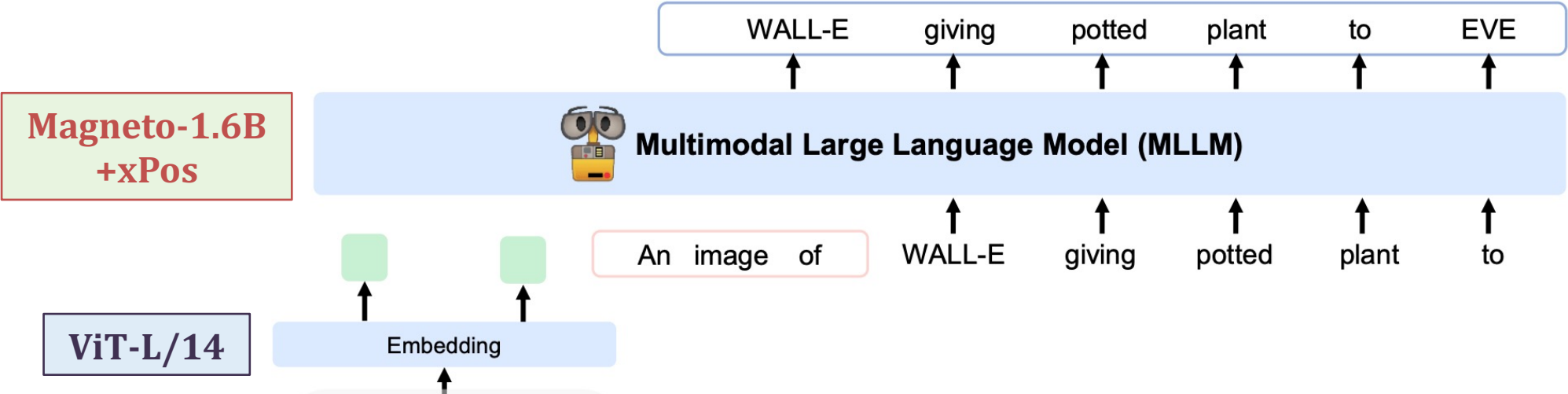
Language Is Not All You Need: Aligning Perception with Language Models

Shaohan Huang*, Li Dong*, Wenhui Wang*, Yaru Hao*, Saksham Singhal*, Shuming Ma*
Tengchao Lv, Lei Cui, Owais Khan Mohammed, Barun Patra, Qiang Liu, Kriti Aggarwal
Zewen Chi, Johan Bjorck, Vishrav Chaudhary, Subhojit Som, Xia Song, Furu Wei[†]
Microsoft

<https://github.com/microsoft/unilm>



KOSMOS-1



Datasets	Format Examples
Text	<s> KOSMOS-1 can perceive multimodal input, learn in context, and generate output. </s>
Image-Caption	<s> <image> Image Embedding </image> WALL-E giving potted plant to EVE. </s>
Multimodal	<s> <image> Image Embedding </image> This is WALL-E. <image> Image Embedding </image> This is EVE. </s>

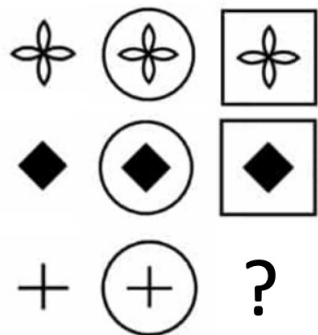
Dataset	Task description	Metric	Zero-shot	Few-shot
<i>Language tasks</i>				
StoryCloze [MRL ⁺ 17]	Commonsense reasoning	Accuracy	✓	✓
HellaSwag [ZHB ⁺ 19]	Commonsense NLI	Accuracy	✓	✓
Winograd [LDM12a]	Word ambiguity	Accuracy	✓	✓
Winogrande [SBBC20]	Word ambiguity	Accuracy	✓	✓
PIQA [BZB ⁺ 20]	Physical commonsense	Accuracy	✓	✓
BoolQ [CLC ⁺ 19]	Question answering	Accuracy	✓	✓
CB [dMST19]	Textual entailment	Accuracy	✓	✓
COPA [RBG11]	Causal reasoning	Accuracy	✓	✓
Rendered SST-2 [RKH ⁺ 21]	OCR-free sentiment classification	Accuracy	✓	
HatefulMemes [KFM ⁺ 20]	OCR-free meme classification	ROC AUC	✓	
<i>Cross-modal transfer</i>				
RelativeSize [BHCF16]	Commonsense reasoning (object size)	Accuracy	✓	
MemoryColor [NHJ21]	Commonsense reasoning (object color)	Accuracy	✓	
ColorTerms [BBBT12]	Commonsense reasoning (object color)	Accuracy	✓	
<i>Nonverbal reasoning tasks</i>				
IQ Test	Raven's Progressive Matrices	Accuracy	✓	
<i>Perception-language tasks</i>				
COCO Caption [LMB ⁺ 14]	Image captioning	CIDEr, etc.	✓	✓
Flicker30k [YLHH14]	Image captioning	CIDEr, etc.	✓	✓
VQAv2 [GKSS ⁺ 17]	Visual question answering	VQA acc.	✓	✓
VizWiz [GLS ⁺ 18]	Visual question answering	VQA acc.	✓	✓
WebSRC [CZC ⁺ 21]	Web page question answering	F1 score	✓	
<i>Vision tasks</i>				
ImageNet [DDS ⁺ 09]	Zero-shot image classification	Top-1 acc.	✓	
CUB [WBW ⁺ 11]	Zero-shot image classification with descriptions	Accuracy	✓	

■ 360B Tokens ■ Batchsize=1.2M (0.5M text, 0.5M image-text, 0.2M interleaved)

Model	VQAv2	VizWiz
Frozen	29.5	-
VLKDViT-B/16	38.6	-
METALM	41.1	-
Flamingo-3B*	49.2	28.9
Flamingo-9B*	51.8	28.8
KOSMOS-1 (1.6B)	51.0	29.2

Model	VQAv2			VizWiz		
	$k = 2$	$k = 4$	$k = 8$	$k = 2$	$k = 4$	$k = 8$
Frozen	-	38.2	-	-	-	-
METALM	-	45.3	-	-	-	-
Flamingo-3B	-	53.2	55.4	-	34.4	38.4
Flamingo-9B	-	56.3	58.0	-	34.9	39.4
KOSMOS-1 (1.6B)	51.4	51.8	51.4	31.4	35.3	39.0

Example of IQ Test

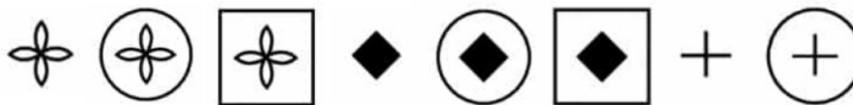


Which option can complete the matrix?

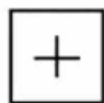


Input Prompt

Here are eight images:

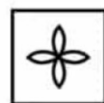


The following image is:



Is it correct?

Yes



Is it correct?

Yes



Is it correct?

Yes



Is it correct?

Yes



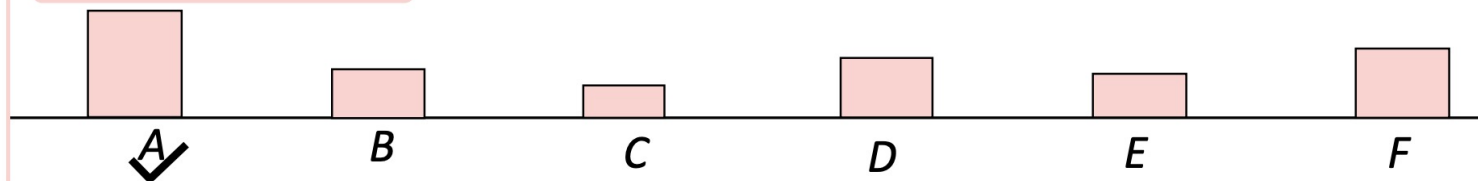
Is it correct?

Yes



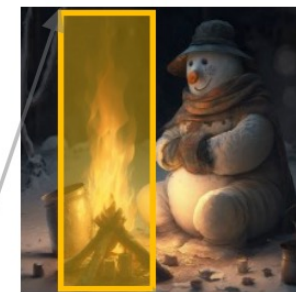
Is it correct?

Yes

 $P(\text{Yes} \mid \text{Input Prompt})$ 

KOSMOS-2: Grounding Multimodal Large Language Models to the World

Zhiliang Peng*, Wenhui Wang*, Li Dong*, Yaru Hao, Shaohan Huang, Shuming Ma, Furu Wei[†]
Microsoft Research
<https://aka.ms/GeneralAI>



Grounding

[a campfire]($\langle loc_4 \rangle \langle loc_{1007} \rangle$)

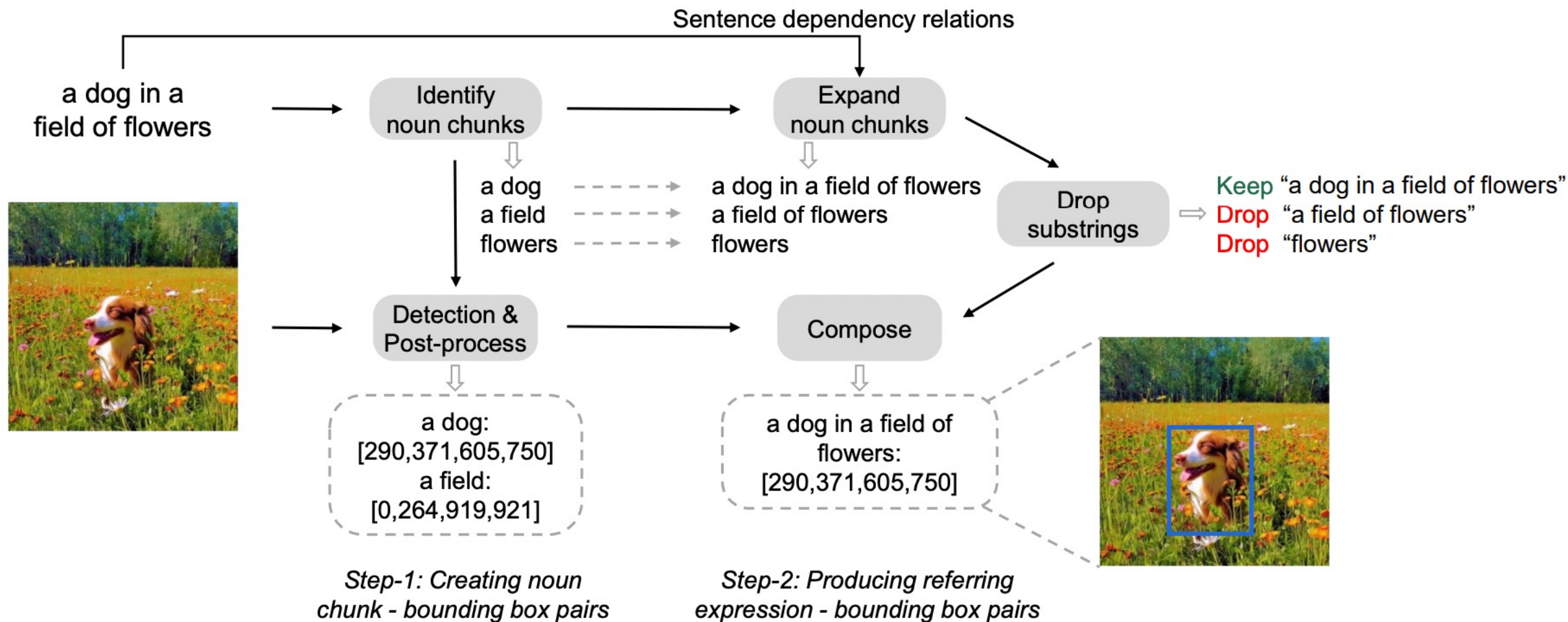
Kosmos-2: Multimodal Large Language Model

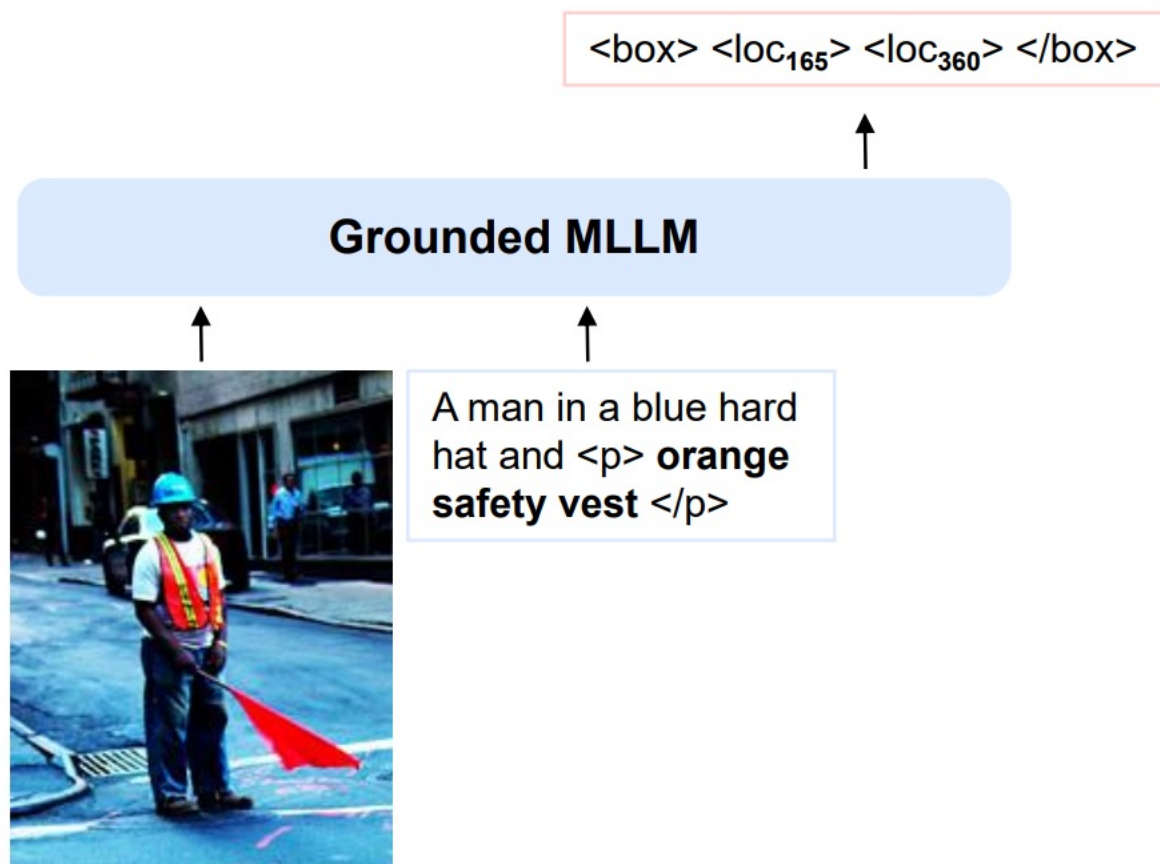


[It]($\langle loc_{44} \rangle \langle loc_{863} \rangle$) sits next to

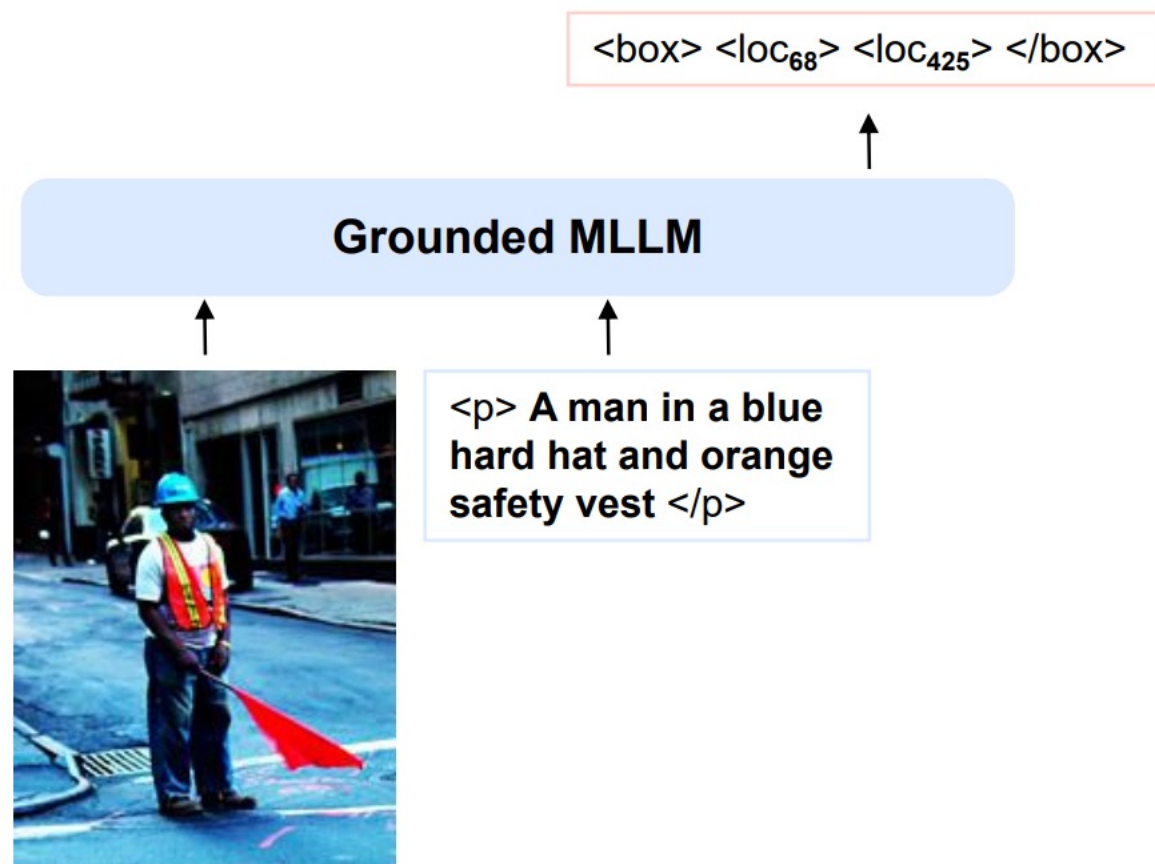


Referring



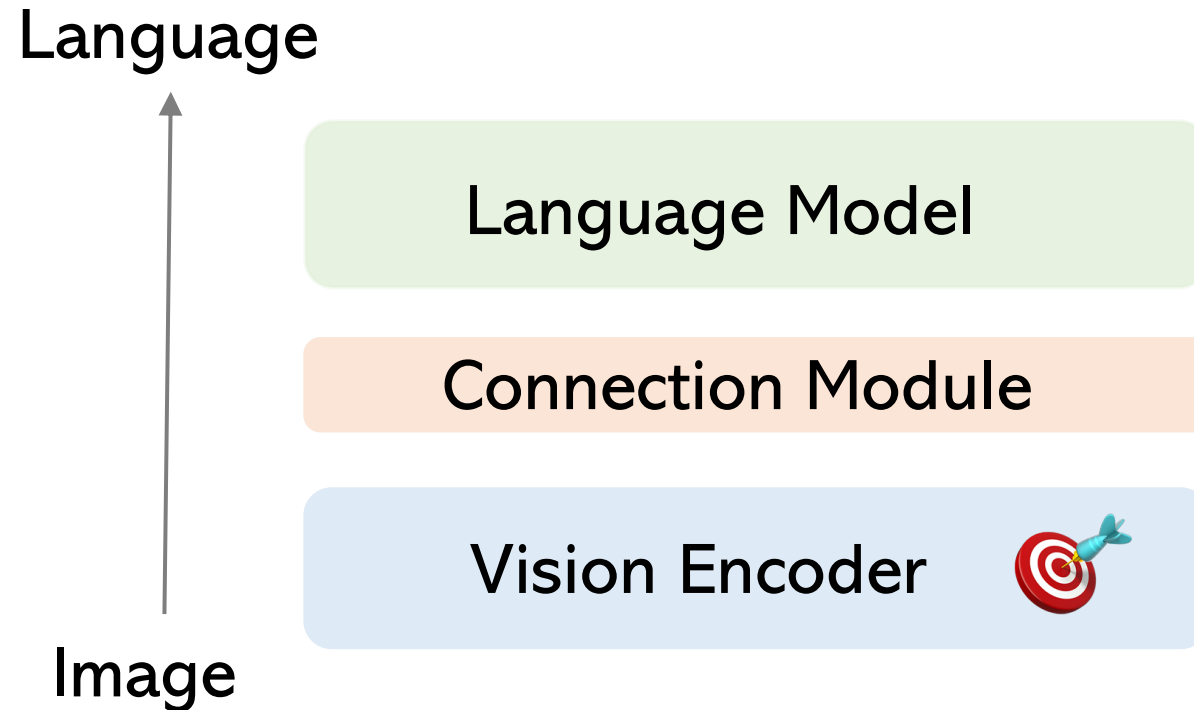


(1) Phrase grounding



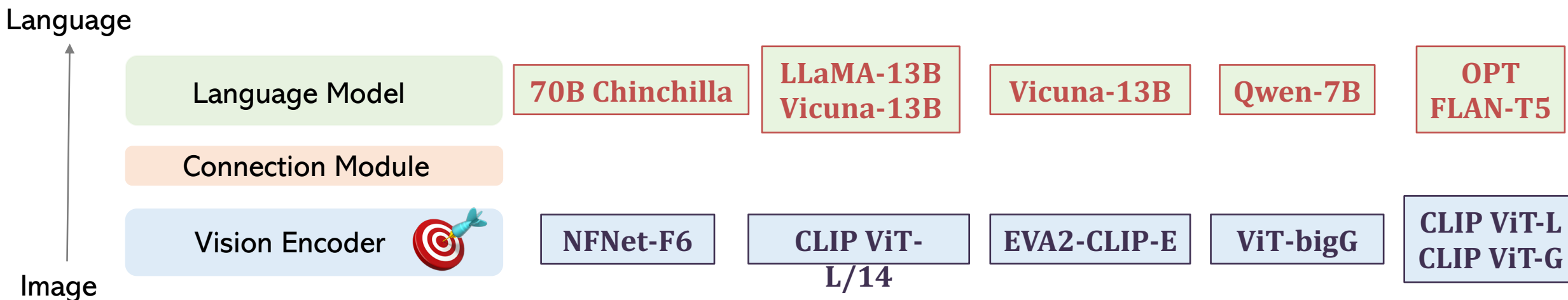
(2) Referring expression comprehension

Q1: Vision Encoder



Q1: Vision Encoder

■ Flamingo ■ LLaVA ■ COGVLM ■ Qwen ■ BLIP*



Q1: Vision Encoder

What's “up” with vision-language models? Investigating their struggle with spatial reasoning

Amita Kamath¹ **Jack Hessel**² **Kai-Wei Chang**¹

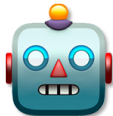
¹ University of California, Los Angeles

² Allen Institute for AI

{kamatha, kwchang}@cs.ucla.edu, jackh@allenai.org

■ CLIP存在空间幻觉

Q1: Vision Encoder

	A dog <u>on</u> a table	A dog <u>right of</u> a table
	A dog <u>left of</u> a table	A dog <u>right of</u> a table
	A dog <u>right of</u> a table	A dog <u>right of</u> a table
	A dog <u>under</u> a table	A dog <u>right of</u> a table

Model	Whats- Up	COCO- spatial	GQA- spatial	Avg
CLIP ViT-B/32	31.0	47.4	46.9	41.8
CLIP ViT-L/14	26.1	49.5	47.3	41.0
NegCLIP	34.4	46.9	46.0	42.4
RoBERTaCLIP	25.1	50.0	49.8	41.6
CoCa	29.4	46.7	47.1	41.0
XVLM 4M	31.5	61.7	58.7	50.6
XVLM 16M	41.9	65.0	58.2	55.0
BLIP 14M	38.5	54.0	49.8	47.5
BLIP 129M	30.4	49.3	49.0	42.9
BLIP2-ITM	37.6	53.0	49.8	46.8
BLIP2-ITC	29.0	53.7	51.0	44.6
FLAVA	30.5	52.6	51.7	44.9
CoCa-Caption	24.1	48.6	49.5	40.8
XVLM-Flickr30K	44.3	65.2	61.4	56.9
XVLM-COCO	42.1	71.0	68.1	60.4
BLIP-Flickr30K	33.8	54.2	48.9	45.6
BLIP-COCO	32.8	51.4	51.4	45.2
BLIP-VQA	47.8	62.0	58.4	56.0
Random / Text-only	25.0	50.0	50.0	41.7
Human Estimate	100.0	97.3	99.0	98.8

What Makes for Good Visual Tokenizers for Large Language Models?

Guangzhi Wang^{1,2*} Yixiao Ge^{2,3†} Xiaohan Ding³ Mohan Kankanhalli¹ Ying Shan^{2,3}

¹National University of Singapore ²ARC Lab, Tencent PCG ³Tencent AI Lab

■ CLIP细粒度理解弱

Q1: Vision Encoder

(a) VQA

(b) Image Captioning

(c) Object Counting

(d) Multi-class Identification

Q: Are the men competing?

Q: What does this image describe?

Q: How many people are there in the image?

Q: Does sports shoe exist in the image?



A: Yes

A: Some males are playing with a white frisbee.

A: 4

A: Yes

Semantic Understanding

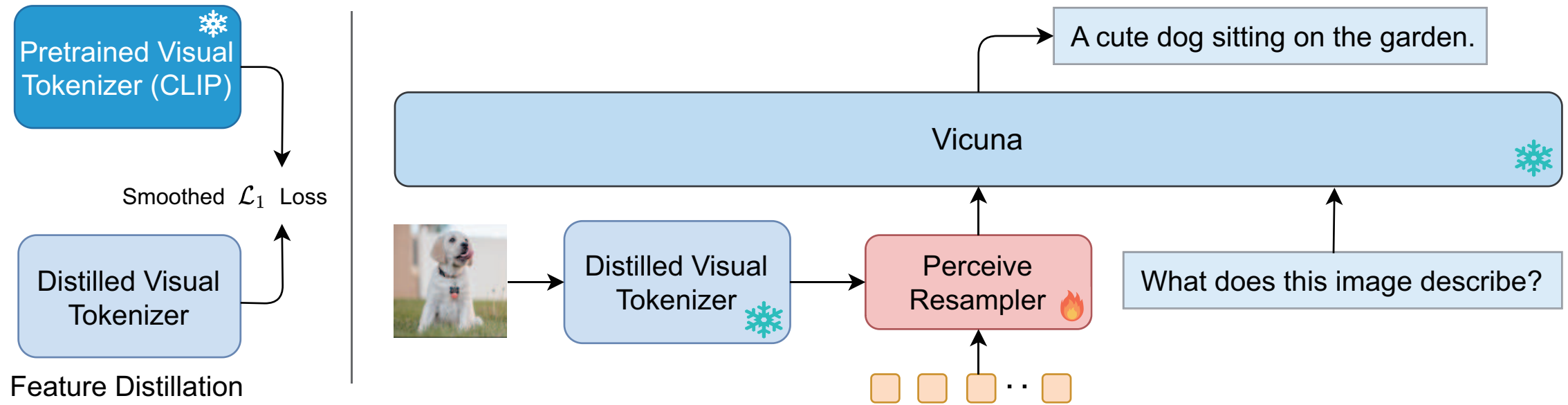
Fine-Grained Visual Perception

Joint Tuning	Supervised	Visual Tokenizer	# Pretraining Images	VQA Acc	Captioning CIDEr	SPICE	OC Acc	MCI Acc	Avg
×	Fully	DeiT [16]	1.28 M	48.3	65.8	15.9	37.5	83.6	58.8
	Self	DINO [19]	1.28 M	50.1	45.0	13.5	46.5	80.8	55.6
		MAE [18]	1.28 M	48.4	37.3	11.8	47.5	82.7	53.4
		DINOv2 [20]	142 M	<u>51.3</u>	<u>67.9</u>	<u>16.1</u>	<u>47.0</u>	86.0	63.1
	Weakly	CLIP [17]	400 M	52.2	69.3	16.6	42.5	86.0	<u>62.5</u>
✓	Fully	DeiT [16]	1.28 M	50.7	38.4	10.0	41.0	86.9	54.3
	Self	DINO [19]	1.28 M	47.3	54.1	14.5	44.5	86.6	58.1
		MAE [18]	1.28 M	48.9	48.0	14.2	47.5	88.7	58.2
		DINOv2 [20]	142 M	50.5	49.6	13.0	43.5	84.1	56.9
	Weakly	CLIP [17]	400 M	47.7	64.2	15.4	45.5	<u>88.0</u>	61.4

Q1: Vision Encoder

Joint Tuning	Visual Tokenizer	VQA Acc	Captioning CIDEr	SPICE	OC Acc	MCI Acc	Avg
×	CLIP [17]	52.2	69.3	16.6	42.5	86.0	62.5
×	RegionCLIP [21]	48.7	28.5	10.3	41.0	86.0	51.5
×	Owl-ViT [22]	44.0	32.5	8.5	43.0	80.8	50.1
✓	CLIP [17]	47.7	64.2	15.4	45.5	88.0	61.4
✓	RegionCLIP [21]	49.7	65.5	14.1	47.5	86.4	62.3
✓	Owl-ViT [22]	50.8	61.2	14.0	38.5	87.1	59.4

Q1: Vision Encoder



Q1: Vision Encoder

Model	#Vis. Tok. Params	VQA Acc	COCO-Caption CIDEr	SPICE	COCO-OC Acc	COCO-MCI Acc	VCR-OC Acc	VCR-MCI Acc	Avg
Flamingo-9B [6]	438 M	51.8	79.4	-	-	-	-	-	-
Kosmos-1 [10]	307 M	51.0	84.7	16.8	-	-	-	-	-
LLaVa [8]	307 M	39.0	48.3	15.0	22.2	52.0	24.6	66.9	44.7
miniGPT4 [7]	1.0 B	58.2	80.6	<u>19.5</u>	21.5	76.8	<u>25.1</u>	<u>70.1</u>	55.4
BLIP-2 [5]	1.0 B	62.4	93.3	17.3	<u>48.0</u>	<u>81.9</u>	20.2	68.9	<u>62.5</u>
GVT (Ours)	307 M	<u>60.4</u>	<u>89.9</u>	19.6	56.2	89.3	40.3	78.9	69.2

Q1: Vision Encoder

1

CL: CLIP及其变种

2

自监督: MAE等

3

下游任务监督: RegionCLIP等

4

厂商私有

5

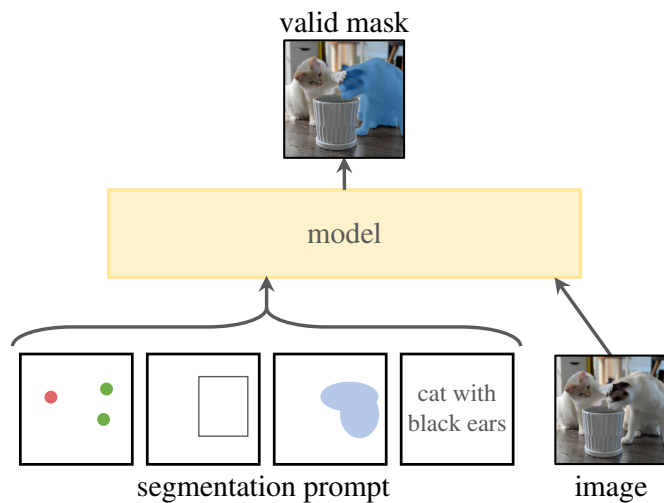
多视角融合

Q1: Vision Encoder-SAM

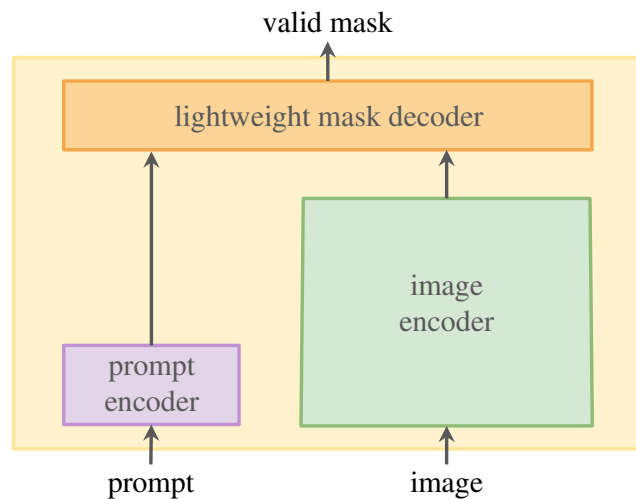
Segment Anything

Alexander Kirillov^{1,2,4} Eric Mintun² Nikhila Ravi^{1,2} Hanzi Mao² Chloe Rolland³ Laura Gustafson³
 Tete Xiao³ Spencer Whitehead Alexander C. Berg Wan-Yen Lo Piotr Dollár⁴ Ross Girshick⁴
¹project lead ²joint first author ³equal contribution ⁴directional lead

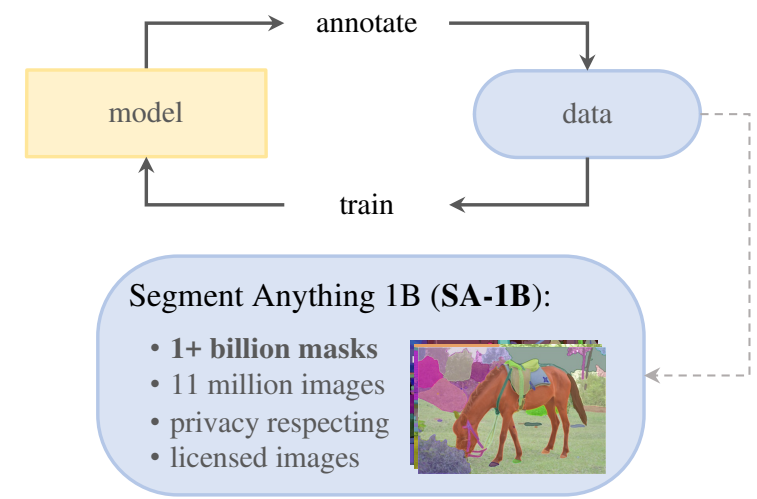
Meta AI Research, FAIR



(a) **Task:** promptable segmentation



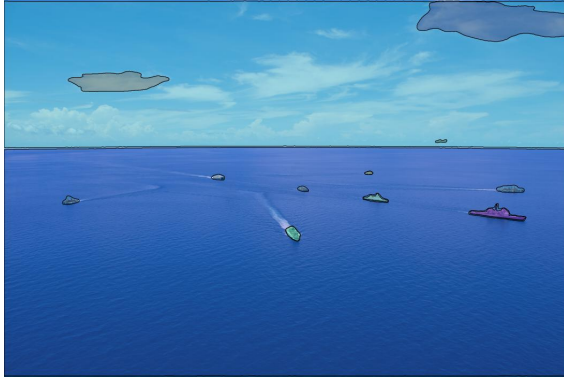
(b) **Model:** Segment Anything Model (SAM)



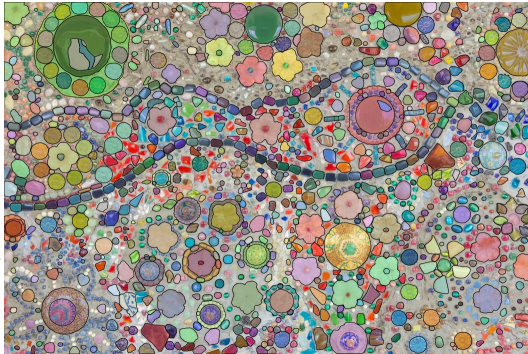
(c) **Data:** data engine (top) & dataset (bottom)

Q1: Vision Encoder-SAM

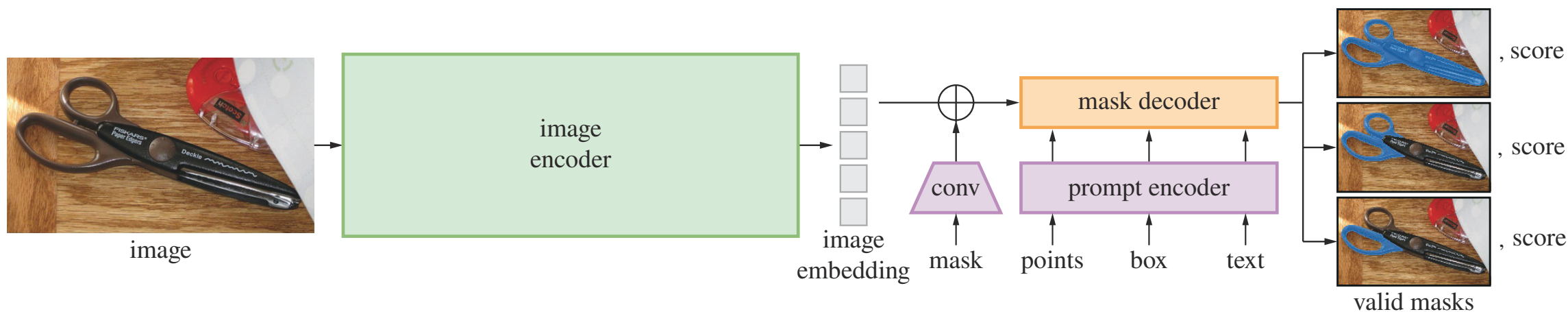
< 50 masks



> 500 masks



Q1: Vision Encoder-SAM



■ <https://huggingface.co/facebook/sam-vit-large>

■ 分辨率 $\rightarrow 1024 * 1024$

■ 输出patch $\rightarrow 64 * 64$

Q1: 复旦眸思 (MouSi)

MouSi: Poly-Visual-Expert Vision-Language Models

Xiaoran Fan*, Tao Ji*, Changhao Jiang*, Shuo Li*, Senjie Jin*,

**Sirui Song, Junke Wang, Boyang Hong, Lu Chen,
Guodong Zheng, Ming Zhang, Caishuang Huang,
Rui Zheng, Zhiheng Xi, Yuhao Zhou, Shihan Dou, Junjie Ye, Hang Yan,**

Tao Gui[†], Qi Zhang[†], Xipeng Qiu, Xuanjing Huang, Zuxuan Wu, Yu-Gang Jiang

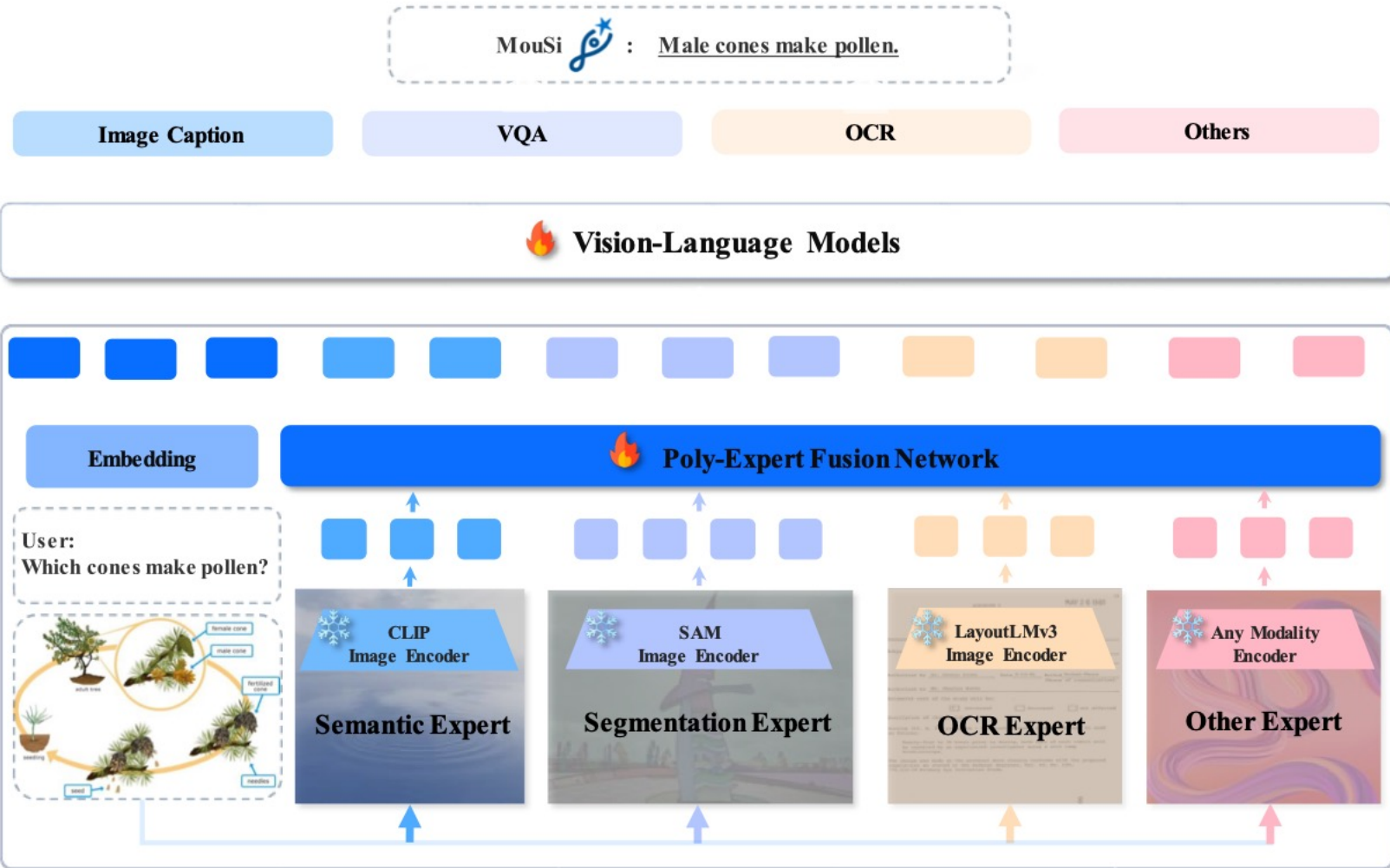
Fudan NLP Lab & Fudan Vision and Learning Lab

Q1: 复旦眸思 (MouSi)

Expert	Res.	Param.	d_hid	#Patch	Type	Pre-training	
						Tasks	Images
CLIP	336	300M	1024	576	ViT	Image-Text Matching	400M
DINOv2	224	1.1B	1536	256	ViT	DINO+iBOT+SwAV	142M
LayoutLMv3	224	368M	1024	196	ViT	Document OCR	11M
ConvNeXt	384	200M	768	1024	CNN	Image Classification	2B
SAM	1024	637M	1280	4096	ViT	Image Segmentation	11M
MAE	224	630M	1280	256	ViT	Patch-level Denoising	1.3M

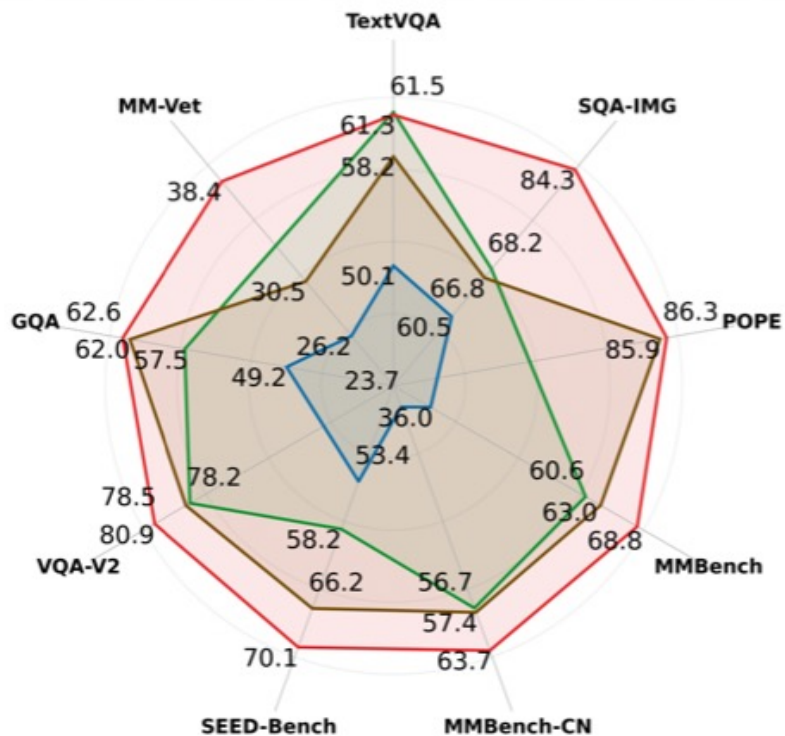
- 不同专家各有所长，是否能专家协同合作？

Q1: 复旦眸思 (MouSi)

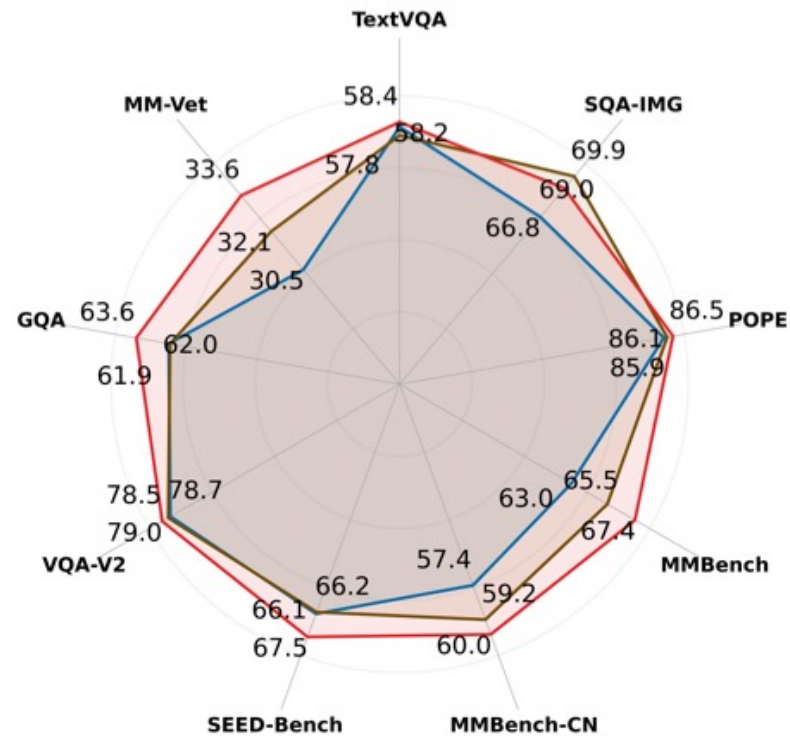


Q1: 复旦眸思 (MouSi)

— InstructBLIP — Qwen-VL-Chat — LLaVA-1.5-7B — MouSi

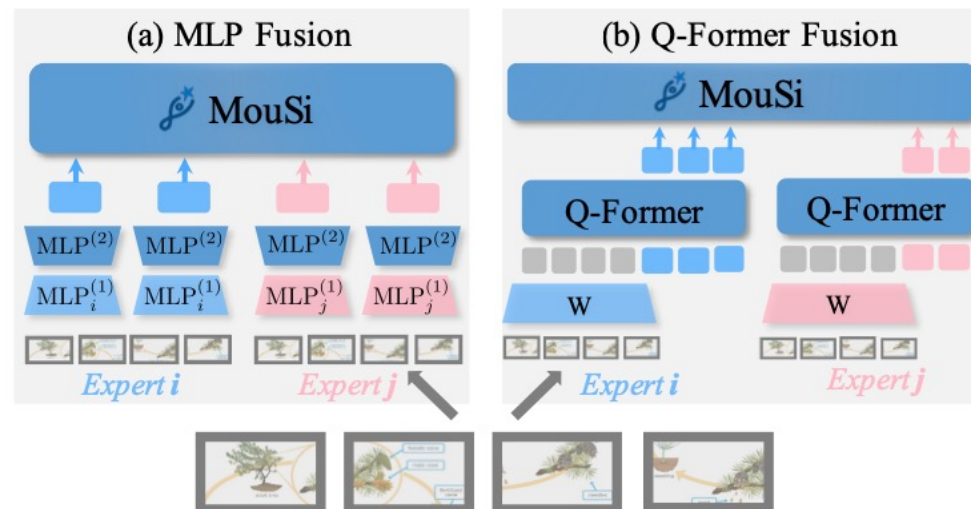


— Best **Single** Expert — Best **Double** Expert — Best **Triple** Expert



Q1: 复旦眸思 (MouSi)

- 连接层设计
- 减少视觉token



- 位置编码
- 避免位置溢出

text₁ text₂ text₃ text₄



Image

5	6	7	8
9	10	11	12
13	14	15	16
17	18	19	20

(a) original

5	5	5	5
5	5	5	5
5	5	5	5
5	5	5	5

(b) share-all

5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

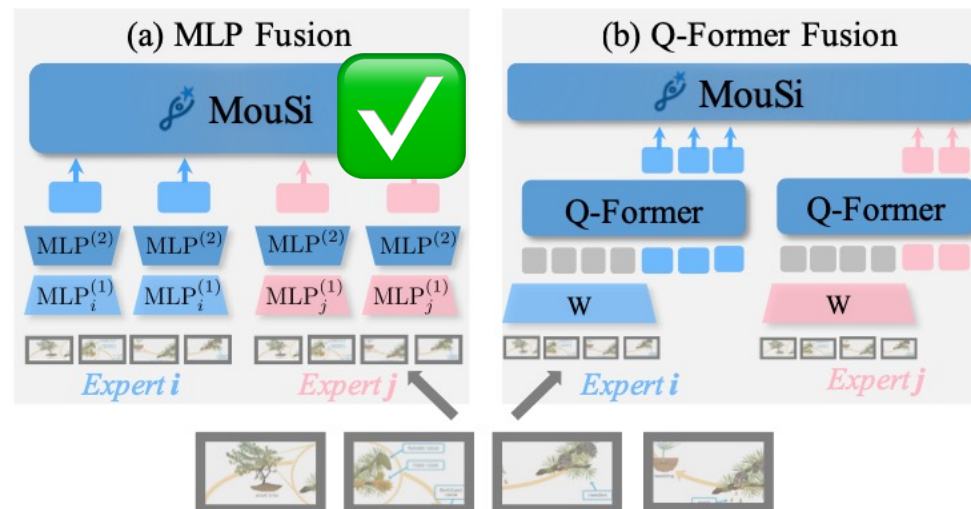
(c) share-by-row

P	PE from LLM			
5	5⊕1	5⊕2	5⊕3	5⊕4
6	6⊕1	6⊕2	6⊕3	6⊕4
7	7⊕1	7⊕2	7⊕3	7⊕4
8	8⊕1	8⊕2	8⊕3	8⊕4

(d) share-by-row&col

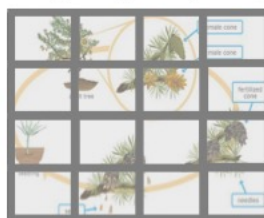
Q1：复旦眸思（MouSi）

- 连接层设计
- 减少视觉token



- 位置编码
- 避免位置溢出

text₁ text₂ text₃ text₄



Image

5	6	7	8
9	10	11	12
13	14	15	16
17	18	19	20

(a) original

5	5	5	5
5	5	5	5
5	5	5	5
5	5	5	5

(b) share-all

5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

(c) share-by-row

P PE from LLM

5⊕1	5⊕2	5⊕3	5⊕4
6⊕1	6⊕2	6⊕3	6⊕4
7⊕1	7⊕2	7⊕3	7⊕4
8⊕1	8⊕2	8⊕3	8⊕4

(d) share-by-row&col

Q1: Apple (MM1)

MM1: Methods, Analysis & Insights from Multimodal LLM Pre-training

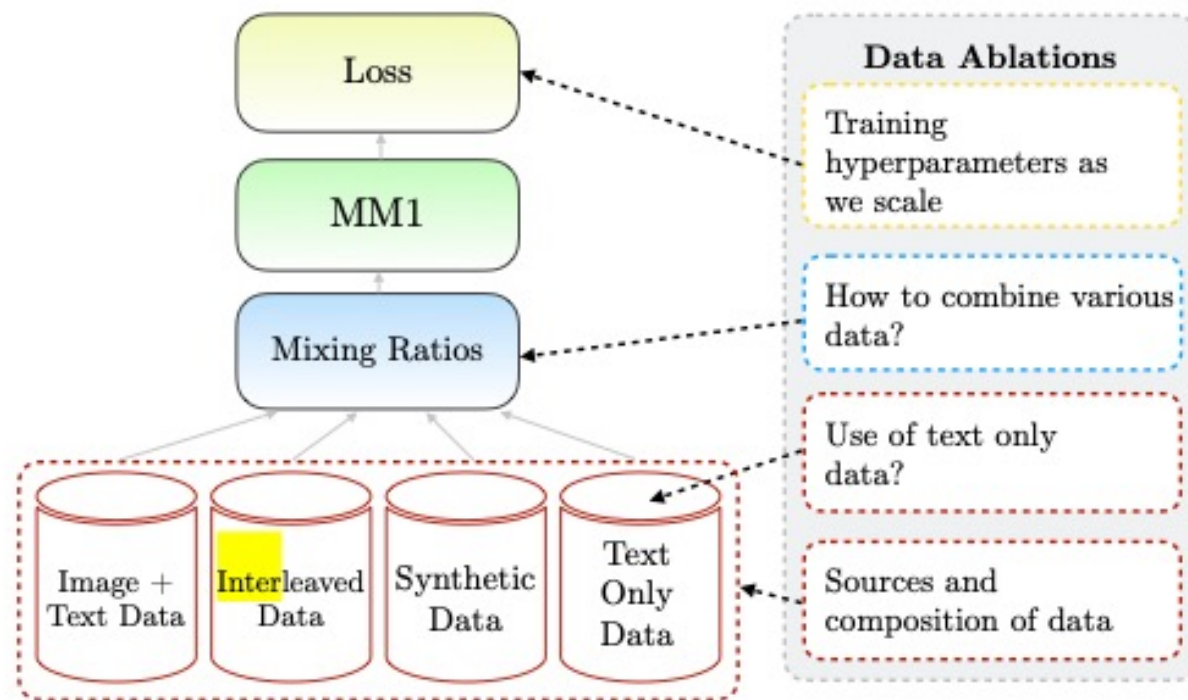
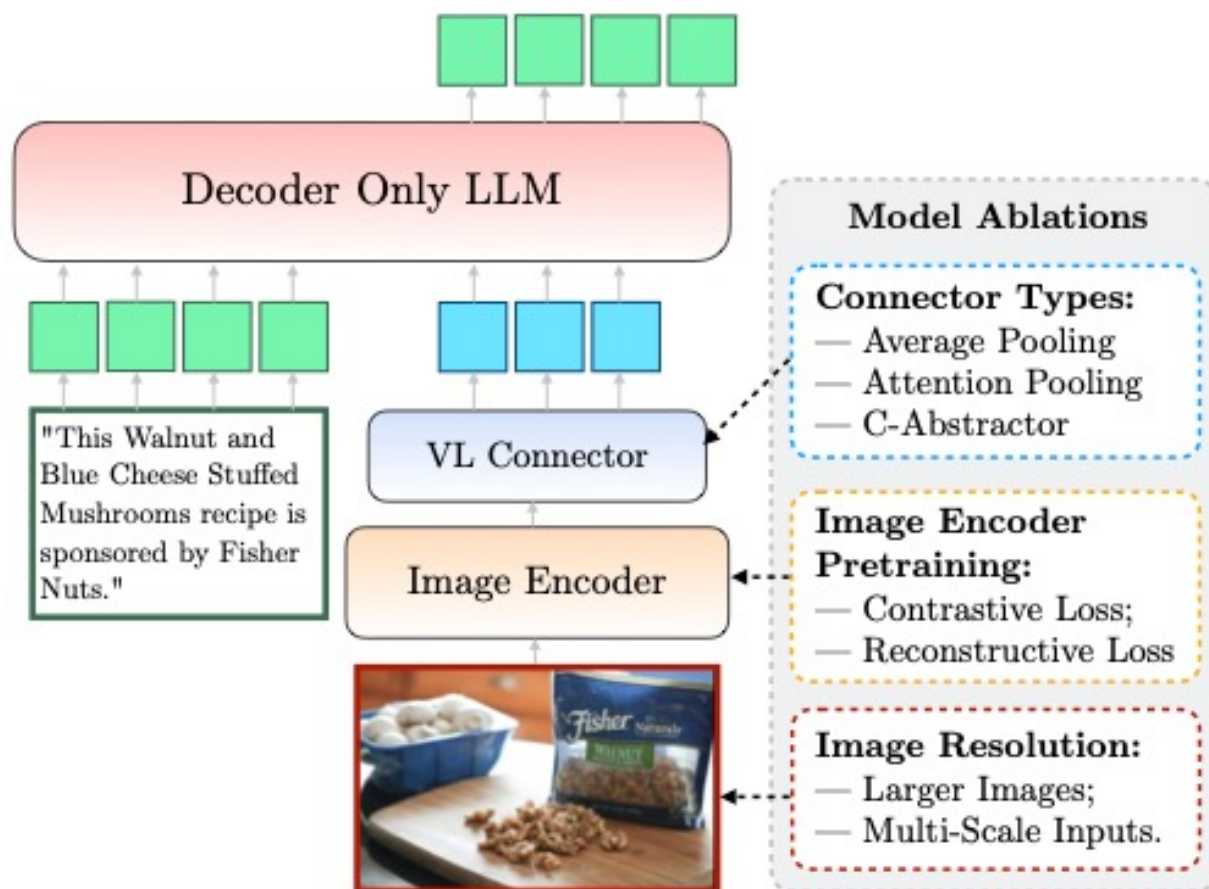
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Q1: Apple (MM1)



Q1: Apple (MM1)

		Setup			Results		
	Model	Arch.	Image Res.	Data	0-shot	4-shot	8-shot
Recon.	AIM _{600M}	ViT/600M			36.6	56.6	60.7
	AIM _{1B}	ViT/1B	224	DFN-2B	37.9	59.5	63.3
	AIM _{3B}	ViT/3B			38.9	60.9	64.9
Contrastive	CLIP _{DFN+VeCap}	ViT-L		DFN-5B+VeCap	36.9	58.7	62.2
	CLIP _{DFN}	ViT-H	224	DFN-5B	37.5	57.0	61.4
	CLIP _{DFN+VeCap}	ViT-H		DFN-5B+VeCap	37.5	60.0	63.6
	CLIP _{DFN+VeCap}	ViT-L			39.9	62.4	66.0
	CLIP _{DFN+VeCap}	ViT-H	336	DFN-5B+VeCap	40.5	62.6	66.3
	CLIP _{OpenAI}	ViT-L		ImageText-400M	39.3	62.2	66.1
	CLIP _{DFN}	ViT-H	378	DFN-5B	40.9	62.5	66.4